



Quad satellite system

Manual v1.0.8

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Safety warning

Before installing this module, ensure your Eurorack case is powered OFF.

Connect the power ribbon with correct orientation to avoid damage to the module or power supply. Use only standard Eurorack $\pm 12\text{V}$ power and avoid exposure to moisture, excessive heat, or direct sunlight. Make sure all signals stay within typical Eurorack voltage ranges (usually $\pm 5\text{V}$ to $\pm 10\text{V}$), and do not connect incompatible sources. Allow for ventilation, use proper mounting screws, and keep the module away from children and pets. Do not attempt repairs or modifications unless you are qualified—unauthorized changes may cause malfunction or void warranty and cause permanent damage to the module and/or your power supply.

Installation

1. Power OFF your system.
2. Connect the power supply ribbon cable in the correct orientation
3. Use the provided screws (or any compatible M3 screws) to mount it on your system.

Specs

height 3U | width 18hp | depth 25mm
power consumption ~190mA- +12V, ~20mA -12V
48khz 16bit

Features

Unique UI
Spectral sampling™
8x spectral tables slots
4x 20 bands filterbanks
4x internal oscs
4x Warp modifiers
based on esp32 + daisy 2
127x127px monochrome OLED display
48khz 16bit audio

Connectivity

6x CV inputs
3x trig/gate inputs
3x audio inputs
1x REC input
4x audio outputs
1x audio sum out

back:
i2c port
expansion port
usb C

Panel Overview

Key
Musically control the tonic of the satellites (1V/oct input)

Orbit
Set the speed of the satellites

Time
offset satellites position

Satellite Selector
Edit ORBIT and TIME for each satellite individually

Set
Edit Orbit start & end points (hold to enter SAVE page)

Record
Hold REC or send a gate to record a new spectral table

REC AUDIO Input
Send audio to be analyzed translated into a Spectral table

Spectral table Selector
Eisei can store and recall 8 spectral tables

Spread
Set the frequency intervals of the Satellites, from unison to octave unison - and anything in between.

Lens
Filterbank shaper.
At Noon all bands are wide open, the spectral table is unaltered. CCW for lopass-like response, CW for spectrum tilt towards the hi end.

Warp Mode
This section provides four methods to further manipulate the signal: Shift, Tanh, Shape and Blur

Reset
Set the satellites to their "0" position keeping the Time offset

All out
analog sum of outs A-D

Input D
activates satellite D's scanner mode

unknown devices

key

orbit

time

spread

lens

SATELLITE

WARP

SET

IN

REC

A

B

OUTS

C

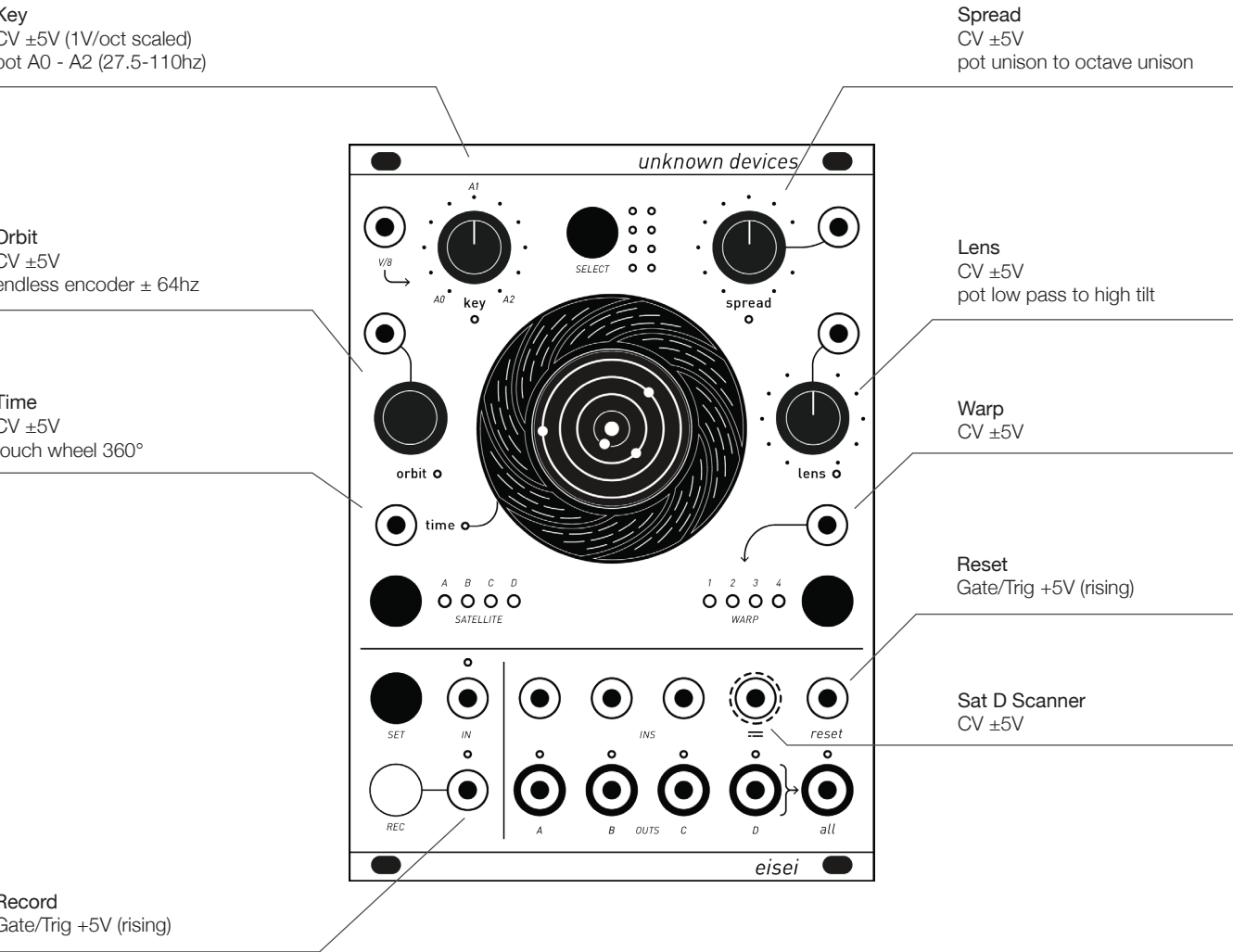
D

all

reset

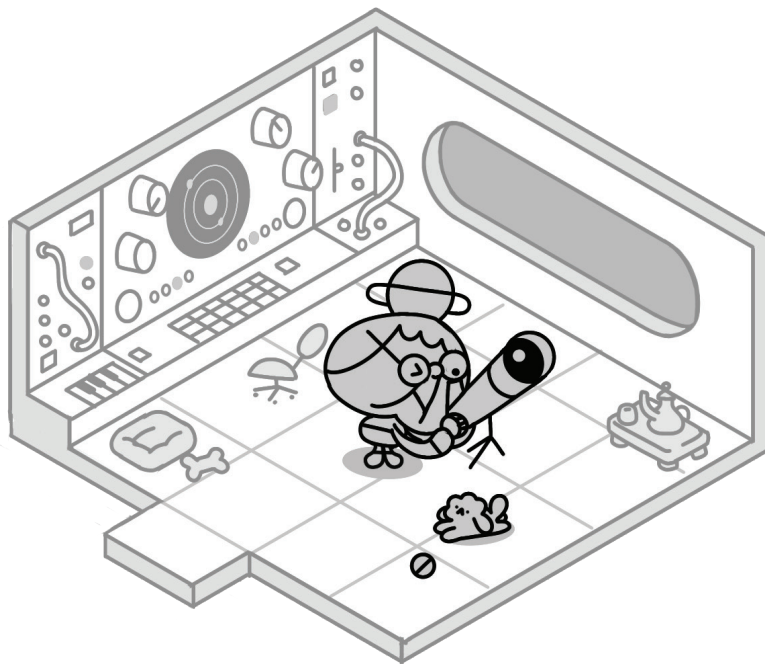
eisei

Parameters ranges (CV & pots)



Story

Granny is a world renowned astrophysicist expert in satellite and analisis tech. Her husband went to space for a mission but never came back. She lives with Planck, her tiny dog, in her observatory. Help Granny find her lost in space husband by operating the quad satellite system Eisei. Receive and process signal from outer space to find any clue!



Characters

Granny

Species: Human

Age: unknown

likes: tea and mochi, summer nights

dislikes: low quality solder, cat people

Planck

Species: Dog

Age: prosumably around 8

likes: a good glass of wine, cosplay

dislikes: any living being except Granny (she's tolerable)

Satellites

In this environment a satellite is a combination of 3 elements:

1. playhead to read the spectral table
2. filterbank to which the spectral table reading is applied
3. sound source, either using the internal pulse wave oscillator (one for each satellite) or external signal using the dedicated input - exception made for input D, more on that later.

Eisei features 4 satellites A, B, C, and D.

Time

This parameter acts as an offset on the position of the satellite on the orbit path. Can be edited manually using the touch wheel and controlled using the dedicated CV input.

- *Global operation*

Touch the Time Wheel or send a CV into the time CV IN to move all the satellites along their orbit path. Positive voltage will offset the satellites clockwise, negative voltage will offset them counterclockwise. The Range 0V to +5V corresponds to a full lap.

- *Individual operation (touch wheel only)*

Use the *SATELLITE* selector button to select the satellite you want to move along the orbit path, the corresponding LED will be lit to confirm selection and the satellite on the screen will be highlighted. Now touch the time wheel will move the selected satellite.

After 5 seconds of inactivity the selection will be disabled and will be back to default (all 4 satellites selected).

Sending a trig or a gate to the reset input resets the position of the satellites maintaining *time* offset.

Shortcut

Long press Satellite button to restore default *time* position for all satellites.

Orbit

Each satellites runs on its own orbit at its own pace. Sweeping through their orbits the satellites read the Spectral table which contains all the spectrum info captured when holding REC. Its possible to set individually set each satellite speed and direction and highlight a slice of interest to loop in (more on this later).

- Global operation (*SPEED*)

Use the orbit encoder to set the orbit speed value, ranging from 0.00 hz to 64 hz in both clockwise (+) and counterclockwise (-) directions.

Control Voltage sent into the orbit cv input multiplies the main frequency in the range 0.1/10x in both directions mapping voltage as follows:

$$0/+5V = 1/10x$$

$$-5/0V = 0.1/1x$$

- Individual operation (*MULT*)

Use the *SATELLITE* selector button to select the satellite you want to edit, the corresponding LED will be lit to confirm selection and the satellite on the screen will be highlighted. Now turn the orbit encoder to engage the *MULT* page to set a multiplier of the main orbit speed with stepped values ranging from x0 to x10 in both clockwise and counter clockwise directions.

After 5 seconds of inactivity the selection will be disabled and will be back to default (all 4 satellites selected).

Orbit settings are persistent across power cycle.

Shortcut

Long press Orbit encoder to restore default orbit *MULT* values (x1).

Push down the orbit encoder to lock the satellites.

Press the encoder once again to unlock the satellites.

Spectral tables

Eisei can record Spectral tables (aka Datum), a particular type of sample that stores 20 sequences of values corresponding to the 20 bands of the filterbanks. It's like 20 streams of automations extracted from the analyzed audio. Eisei can store up to 64 Spectral Tables, a table can be up to 12 seconds long.

Recording

Select a slot using the SELECT button, the corresponding LED will be lit.

Plug some audio signal into the rec audio in.

Press & Hold the REC button or send a gate to the corresponding gate input to start recording, the corresponding LED will light up red.

Recording will start as soon as the button is pressed or a high gate is received and will stop as soon as the button is released or the gate is low. Right after that the new spectral table will be available in the slot to be explored.

ATTENTION:

The newly recorded spectral table will temporarily overwrite whatever is on the slot but won't be permanently stored until it is saved. This is to avoid unwanted overwriting and preserve your data.

To SAVE the spectral table hold SET and confirm the procedure (more in the "Saving" paragraph).

Loading tables

Push SELECT button until the desired slot is highlighted.

Hold SELECT to enter the spectral table list.

While in this page use the *orbit* encoder to scroll through the list and use the *time* wheel to scroll through a preview of the highlighted table.

Push the encoder to load the selected table into the current slot.

Slot assignment is persistent and doesn't need saving.

Default Spectral tables

Eisei comes with a handful of useful computer-made spectral tables, ranging from basic test-like sweeps to chaotic madness.

- 1. OddEven - Crossfade between Odd and Even bands.*
- 2. Sinemove - Waving sine animation on bands gain.*
- 3. Pulsar - A single band sweep, full cycle. Similar to a very tight bandpass.*
- 4. Mirror - Symmetric band oscillations.*
- 5. Comb - alternating, evenly spaced peaks.*
- 6. Chirp - Three overlapping Gaussian sweeps.*
- 7. Resonance - Two Gaussian peaks opening and closing.*
- 8. Harmonics - Harmonic series over a fundamental.*
- 9. Cascade - Bands fill up low to high.*
- 10. Scatter - Timbrically interesting Random applied to bands gain.*
- 11. Binary - Per-frame bit pattern, each bit a band.*
- 12. Perlin - Perlin noise movement applied to bands gain.*
- 13 Fractal - Self-similar repeating gaps.*
- 14. VCA - Uniform bands on a triangle envelope. Meant as Amplitude modulation.*

NOTE

You can get rid these spectral tables by plugging Eisei's SD in a computer and manually deleting them.

Saving

Save up to 64 Spectral Tables, a table can be up to 12 seconds long.
Spectral tables are volatile until they are saved.

Hold SET to enter saving mode.

You will be prompted to confirm the procedure.

Make a clockwise circle on the touchwheel to confirm, counterclockwise to cancel.

Eisei will display SUCCESS once the procedure is complete.

If an ERROR screen appears please check the SD on the back, you might have reached memory limit.

New tables are named "preset" followed by 3 digit number, starting from 001 and increasing for each new saved table (eg. the first saved table will be preset001, the second will be preset002 and so on).

Recorded but not yet saved tables will appear in the list as "unsaved preset" and will be lost when turning off the system.

Saved tables are persistent and can be backed up on a computer or any device that can handle a micro SD.

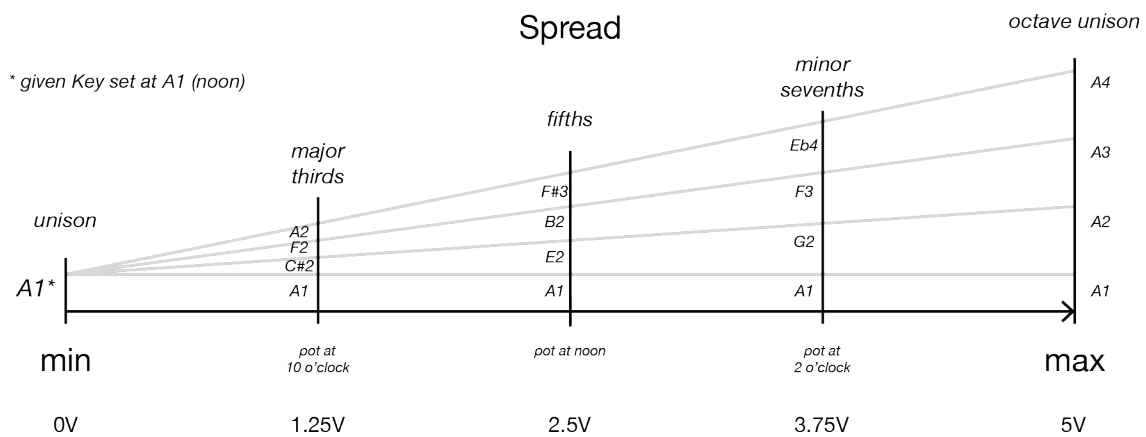
It's also possible to open the micro SD on a device and rename the spectral tables.

Key

The key pot acts just like a tuning key to set the tonic of Eisei, it affects both the internal oscillators and the filterbanks.

The available tuning range for the pot is two octaves, spanning from A0 to A2 (27.5hz - 110hz). This limitation is intentional to best fit the filterbanks response.

The key CV input is 1V/octave scaled. Using the CV IN allows to significantly widen the frequency range of the oscillators from sub audio oscillations to ear pearching high frequencies.



Spread

Satellites are deeply entangled one to each other and move like a hive mind thanks to the Spread parameter. This parameter coordinates the interval between the satellites frequencies. Fully CCW they are be unison, fully CW unravel octave unison, anything in between will unfold happy accidents.

Satellite A is not affected by spread, and sort of acts as reference point to define the tonic, satellites B to D will build on that base frequency each doubling the previous frequency, so for example if key is set to 100hz and spread is fully CW then A will be on 100hz, B on 200hz, C on 400hz, D on 800hz.

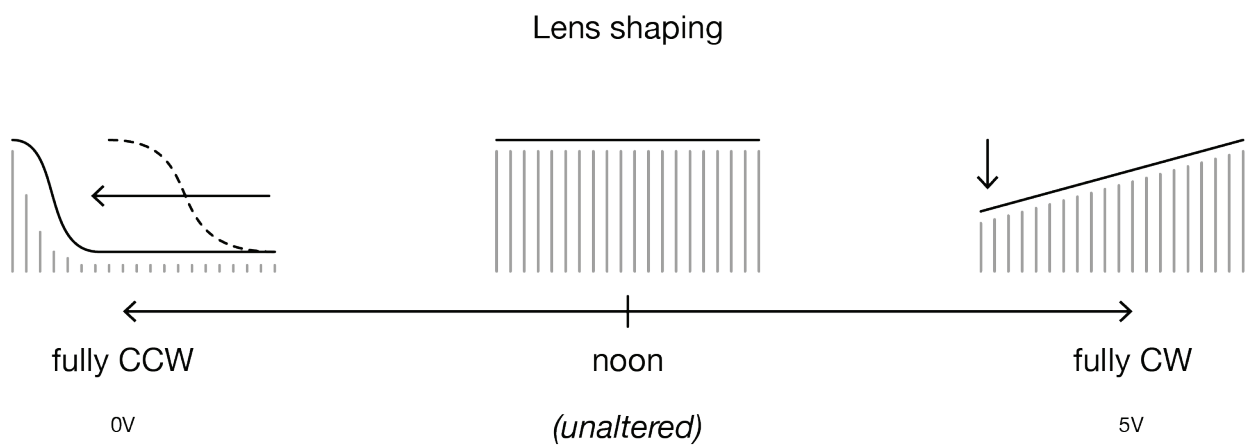
Sending negative voltage might bring to uncharted harmonic territories.

Lens

Lens is all about framing, Classic spectral processor's per band control is re-imagined as a centralized shaper that affects all the filterbanks.

At Noon it's panfocus, all bands are wide open, the spectral table is unaltered.

Sliding counterclockwise smooths out harsh spikes rewarding us with a lowpass like response, while taking the opposite path tilts the spectrum towards the hi end, emphasizing chirp and hisses, without completely cutting the lows.



Warp

This section provides four methods to further manipulate the signal.

1. SHIFT

shifted clone of the bands configuration (post lens)

2. TANH

from gentle distortion to brick clipping

3. SHAPE

internal oscillators pulse width

4. BLUR

bands decay, from short and snappy to smooth and ethereal

Press the Warp button to enter the dedicated page, the corresponding LED will light up blue, and touch the wheel to set the value of the selected warp.

CV Target

Hold the Warp button to set the selected Warp as target of the warp CV input, the corresponding LED will change to warm yellow to confirm and will remain lit. Any combination of Warps can be targeted.

Warp settings are persisted across power cycles.

Warp Animation

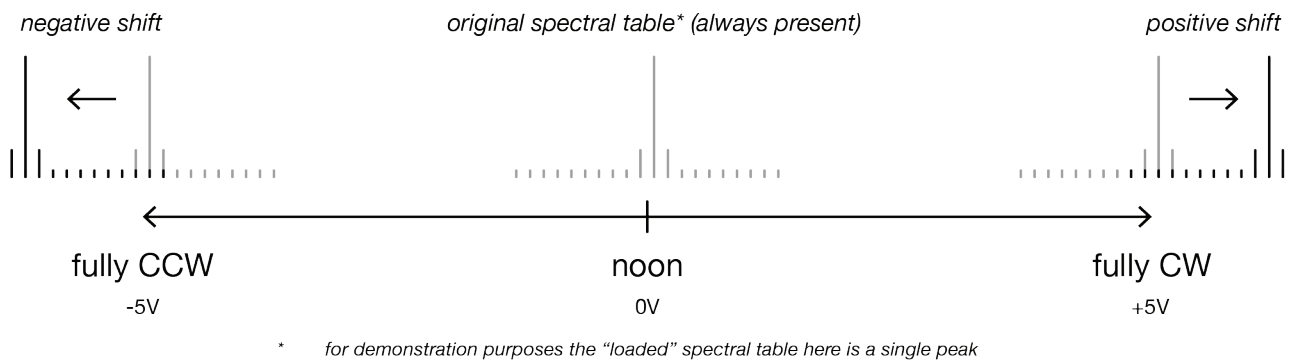
abstract representation of selected warp. Affected by CV

Slider

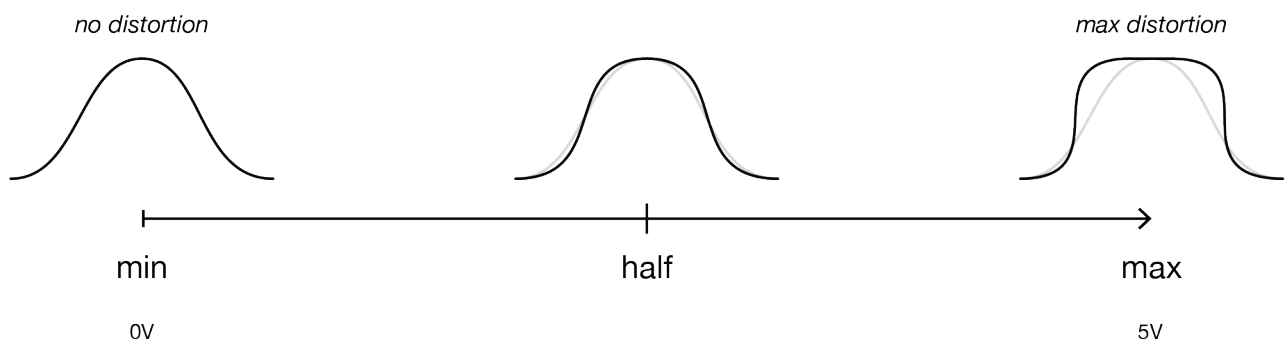
visualizes the offset value. Unaffected by CV



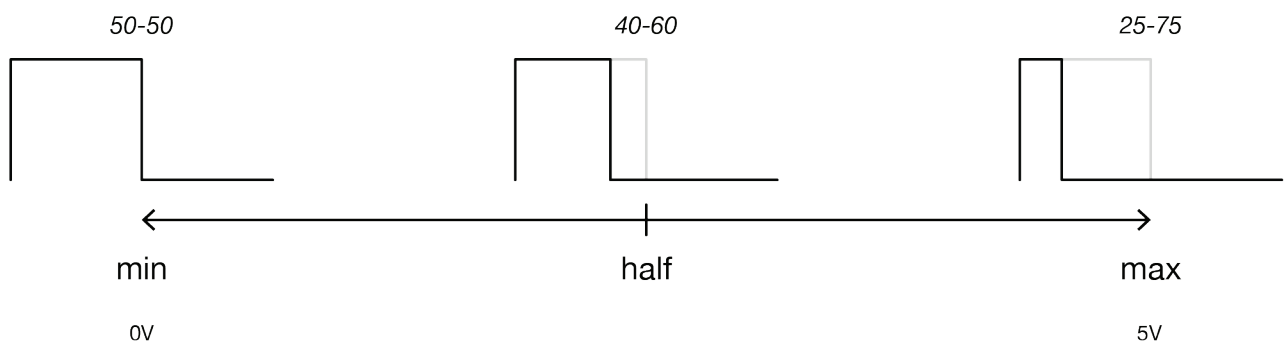
Shift



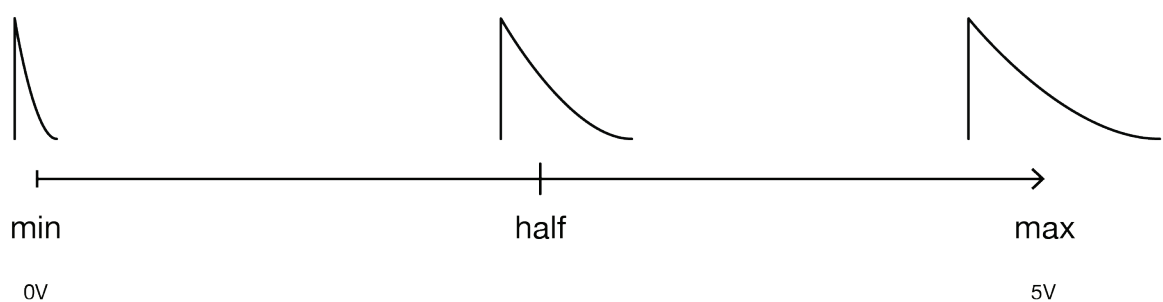
Tan(h)



Shape



Blur



Audio in & outs

Audio inputs

Each satellite input can receive audio except for the satellite D input. Input D can still process the internal oscillator and any audio signal coming from the normalized connections with the other satellites (see scanner mode paragraph below).

NORMALIZATION

Normalization flows left to right, A to D, so a signal plugged into input A will be copied to the following input until it reaches input D or it encounters a plugged cable.

Configuration example:

Stereo to quad - plug the L channel of a stereo signal into input A and the R channel into the input C. the L channel will be processed by both A and B satellites and the R channel will be processed by both C and D satellites.

It is also possible to mix internal oscillators with external audio, for example leaving input A empty but filling the other inputs.

Yes, inputs can be pinged.

SCANNER MODE

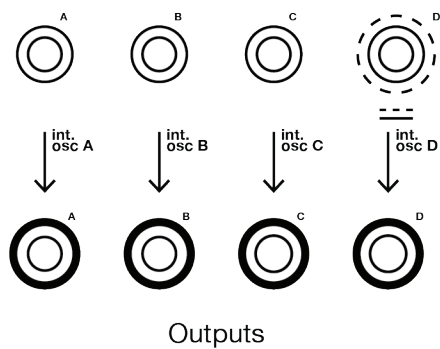
Insert a cable in the input D to enable *scanner* mode. This is satellite D's special ability to crossfade between its own internal oscillator and other satellites audio signals.

When in scanner mode, sat D phase is inverted causing subtraction of sat D signal from the "all" output.

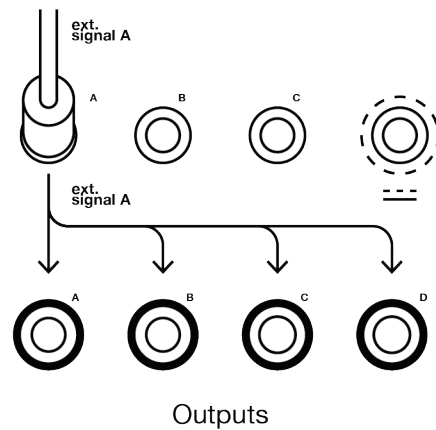
Audio outputs

Each satellites has its dedicated audio output. The *all* out is an analog sum of every satellites output. The all out signal sums each satellite out divided by 4 in amplitude and applies a soft tanh distortion to avoid clipping.

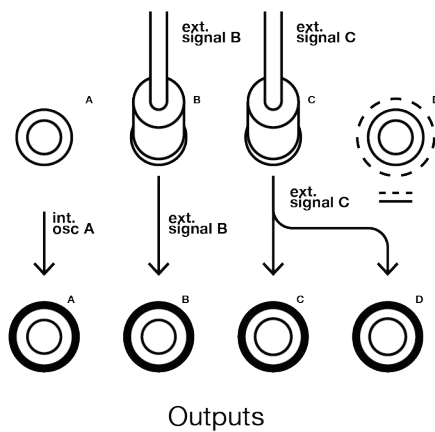
nothing plugged in



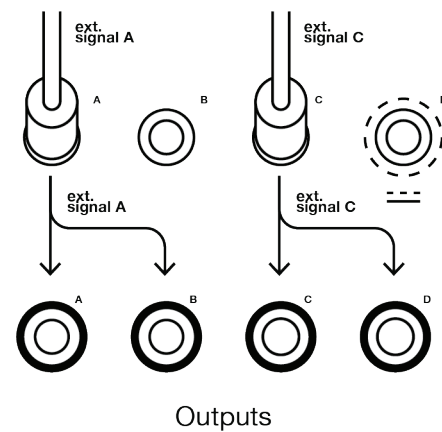
mono to quad



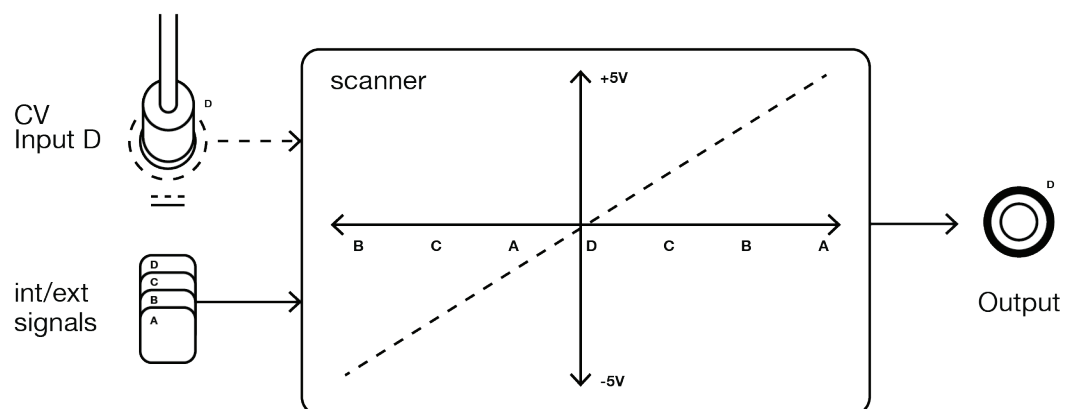
internal + external mixed



dual mono to dual stereo



Sat D Scanner mode



Structure (1)

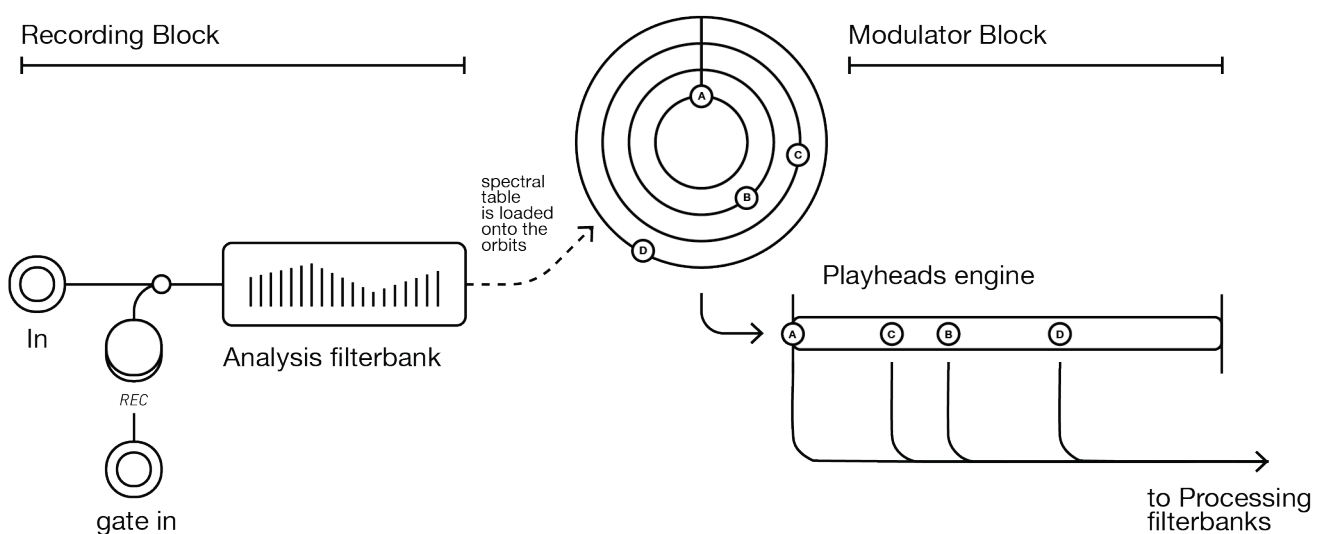
Eisei can be divided in two main blocks:

1. Analysis

This block analyzes any incoming signal and converts it into automations, sort of translating the signal from time domain to frequency domain.

This operation involves an “analysis” filterbank identical to the processing ones but its only purpose is to split the signal into bands and for each band an envelope follower moves according to the amplitude of the band.

The spectral table is the recording of these envelope followers movements.

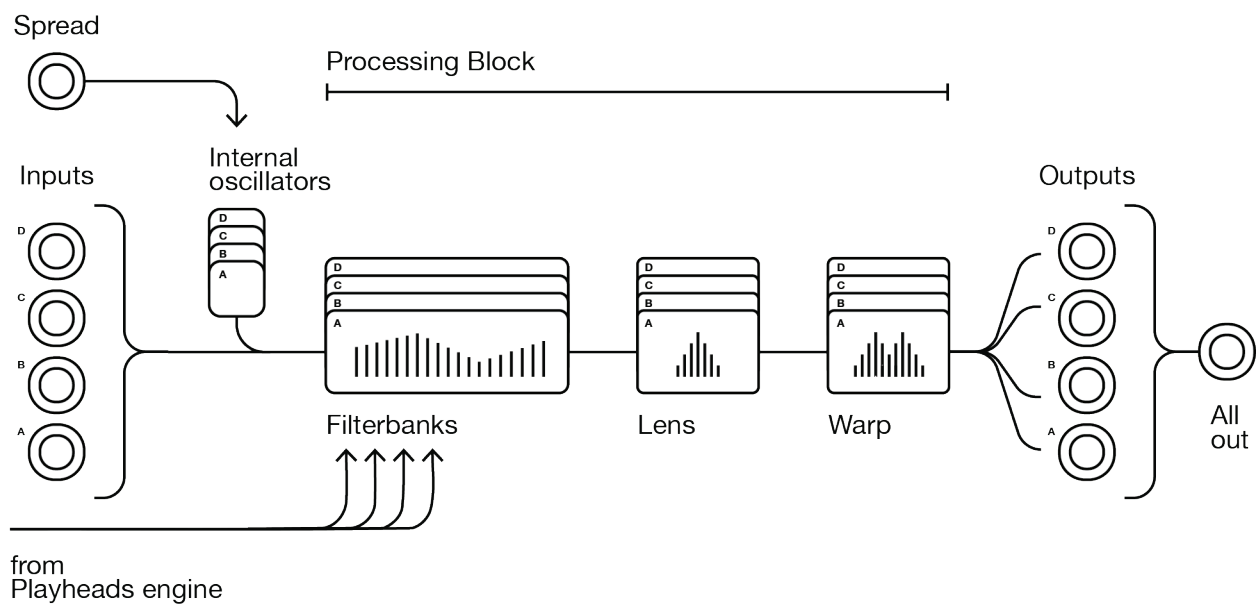


Structure (2)

2. Processing

In this block each satellite audio signal (either internal oscillators or external signal) passes through its filterbank, shaped by the spectral table and lens.

The signal is then processed through the Warp modifiers and finally comes out of the corresponding output.



Button Chart

<i>BUTTON</i>	<i>PRESS</i>	<i>HOLD</i>	<i>HOLD AT BOOT</i>
select	select slot	spectral table list	calibration procedure
set	start/end point	save (confirm with wheel)	advanced settings
rec	record	record	display recover
satellite	select satellite	reset time offsets	
warp	select warp	assign target	
orbit encoder	lock/unlock orbit	reset orbit mults	

Troubleshooting

Eisei doesn't boot.

Probably related to an incomplete firmware update procedure, reflash the firmware.

Unresponsive touch wheel / encoder and no audio.

Probably related to an incomplete firmware update procedure, reflash the firmware or get in touch with us.

The recorded Spectral table disappear after power cycle.

To permanently store a datum it needs to be saved, hold SET and confirm the procedure. Check "saving" paragraph.

Firmware update procedure

Go to [DROP](#) and select the latest firmware (or the desired one) on the right side of the website the tool proposes 2 methods for updating Eisei

The screenshot shows the DROP website interface for firmware updates. At the top, there is a 'version' dropdown menu set to 'v1.0.8' with a red arrow pointing to it, and a 'refresh' button. Below this, the version details for 'v1.0.8' are listed: 'steadier time cv, reliable saving & startup version display' and 'Improvements'. The main content area is divided into two sections: 'Method A (microSD)' and 'Method B (USB C)'. Method A instructions state: 'no power needed - pull the microsd from the module's back, copy both files to the root, reinsert. one click per file, no zip.' It lists 'daisy' as 'firmware.bin' and 'esp32' as 'eisei-1.0.8.esp'. Method B instructions state: 'plug in usb and pick the serial port - the esp32 enters bootloader automatically. the daisy firmware ships alongside (littlefs) and is forwarded to the daisy on next boot.' It shows a 'flash esp32 (serial)' button and a progress bar for 'flash daisy (dfu) - fallback' at 100%.

<https://unwvdevices.github.io/DROP/>

Method A (micro SD)

Eject the micro SD from the side of Eisei then follow these steps:

1. Download both firmware files (daisy + esp32)
2. Copy both files in the root of Eisei's micro SD.
3. Insert the micro SD back into Eisei and power on the system.
4. Eisei will boot and automatically start the update procedure.
5. Once it's done turn off the system, wait a few seconds and restart the system.

The screenshot shows the 'download for microsd' section of the DROP website. It contains the following text: 'no power needed - pull the microsd from the module's back, copy both files to the root, reinsert. one click per file, no zip.' Below this, it lists 'daisy' as 'firmware.bin' and 'esp32' as 'eisei-1.0.8.esp'. Red arrows point to these two links.

Method B (USB C)

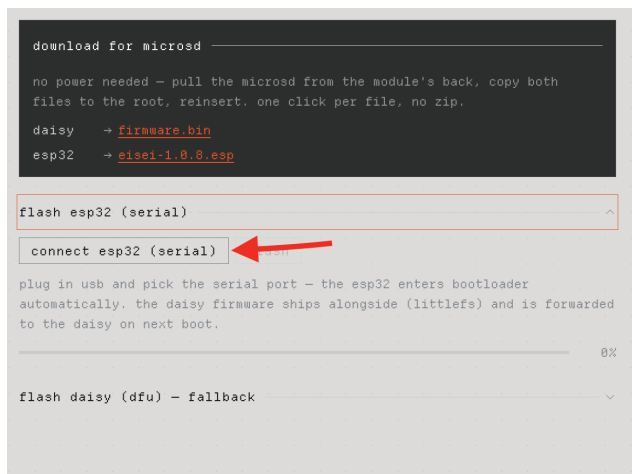
this method is preferred if your unit is running v1.0

!! The module needs to be powered !!

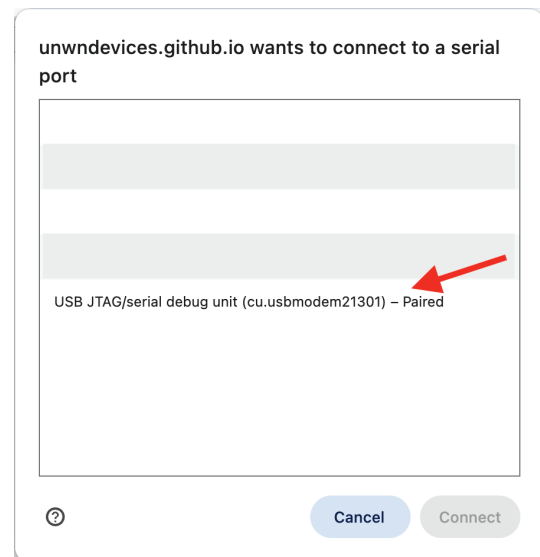
Reach the USB C on the back of the module and use any USB C cable to connect your PC to Eisei, then follow these steps:

1. Click on connect esp32 (serial)
2. A pop-up window will open, select USB JTAG/serial debug unit (it's the ESP32 we can't change that name unfortunately)
3. Click on FLASH and wait for the progress bar to complete
4. Reboot the unit (turn off, then turn on) Eisei will finalise the update showing an update progress bar.

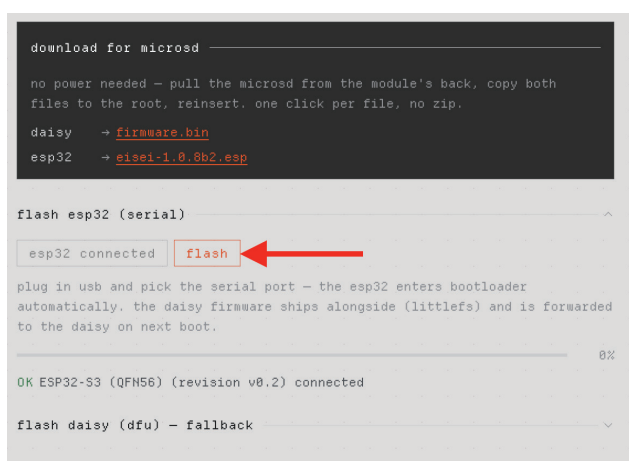
1.



2.



3.



Calibration

All Eisei units are shipped after calibration but if you need to calibrate it differently or the unit lost its calibration you can start the calibration procedure following these steps:

1. turn off your system
2. hold SELECT button
3. turn on your system while holding SELECT

Now Eisei will boot in calibration mode and will display some actions needed to calibrate the module:

1. turn all pots counter-clock wise, then press select to confirm
2. turn all pots clock wise, then press select to confirm
3. unplug all cv, then press select to confirm
4. plug a 1V signal into key input, then press select to confirm
5. plug a 5V signal into key input, then press select to confirm

You're all set!